

IDS2Eval Scorecard

ton-iot-official

17 ok · 2 warning · 3 flag · 22 checks

REVIEW REQUIRED

dedup_check	train duplicates: 0 (0.00%); test rows leaking a train feature-match: 0 (0.00%); test-internal duplicates: 0	ok
label_conflict_check	no feature vector maps to more than one label	ok
near_duplicate_class_check	no near-zero-distance feature vectors found across 10 classes tested	ok
leakage_screen	no single feature or pair dominates importance (top1=13.8%, top2=24.5%)	ok
one_rule_check	single rule 'src_ip <= 3.5' reaches 31.7% test accuracy (30.5% train), vs. 22.7% from always predicting the majority class	ok
feature_auc_ranking_check	'src_port' alone reaches AUC 0.952 identifying 'dos'; a single feature this separable is worth checking for a leakage artifact	flag
identity_column_flag	standalone AUC by column (seen-value coverage, AUC on seen values): src_ip: 0.911 (100.0%, 0.911), dst_ip: 0.891 (99.8%, 0.891), src_port: 0.856 (96.6%, 0.86), dst_port: 0.924 (99.9%, 0.924). Suggest dropping: ['src_ip', 'dst_ip', 'src_port', 'dst_port']	flag
port_protocol_shortcut_check	standalone AUC of 'src_port' + 'proto' combined: 0.865. Port and protocol together may predict the label without attack behavior	flag
temporal_leakage_check	no schema.timestamp_column configured	ok
temporal_realism_check	no schema.timestamp_column configured	ok
flow_group_leakage_check	no schema.flow_id_columns configured	ok
row_order_leakage_check	adjacent-row label-transition rate (train=1.00, test=1.00) as a share of what a randomly shuffled ordering would produce	ok
homogeneity_test	10 classes tested; inconclusive (95% CI straddles the +/-0.05 equivalence margin) for: ['dos']; sample size isn't enough to tell for these classes	warning
resplit_falsification	random-split accuracy=0.9951, grouped-split accuracy=0.9947 (drop=+0.0004); grouped split reproduces random-split accuracy, consistent with inherent class homogeneity rather than a split-artifact explanation	ok
scenario_holdout_falsification	no schema.scenario_column configured	ok
class_distribution_report	10 classes, train imbalance ratio (majority:minority) = 41:1; classes below 1% of train: ['mitm']	warning
low_cardinality_warning	unique values by column: {'src_ip': 51, 'dst_ip': 699, 'src_port': 25548, 'dst_port': 2013}	ok
schema_fingerprint_check	no known extractor signature matched	ok
data_integrity_check	no missing labels, constant features, or +-inf values found	ok
result_robustness_check	accuracy ranges from 0.9759 (identity_columns_dropped) to 0.9951 (random_split) across 4 conditions, a spread of 0.0192 (sampling margin +/-0.0016)	ok
known_issue_lookup	no curated known issues for dataset 'ton-iot-official'	ok
repeated_seed_falsification_check	grouped-split accuracy drop across 10 seeds: mean=+0.0001, 95% CI=[-0.0003, +0.0006]; not material: even the most generous estimate in the 95% CI (+0.0006) stays below the +0.0100 materiality threshold	ok

Class distribution (log scale)



Top feature importances (leakage_screen)

